Meta-WebGuard: An Adaptive Hybrid Framework for Cross-Domain Web Attack Detection

Project Domain: Machine Learning
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Abstract

- Implemented **Meta-WebGuard**, a multi-layered hybrid framework combining Calibrated SVM, Stacked Bi-LSTM, and an XGBoost Meta-Learner.
- Integrated a **Titan-98 Global Mix** dataset, bridging detection gaps using real-world "unseen" samples and CAPEC-aligned injection data.
- Developed a **Stacked Bidirectional LSTM** architecture to capture long-range sequential dependencies in decoded URL strings.
- Achieved high ensemble accuracy by fusing **Shannon Entropy**, brand-spoofing logic, and calibrated model probabilities.
- Optimized for production using **cost-sensitive learning**, applying 2.5x weighting to malicious samples to minimize false negatives.

Introduction

- **Evolving Threat Landscape:** Modern attacks use sophisticated obfuscation to bypass standard filters; our system uses `urllib` decoding to expose hidden payloads.
- **Limitations of Rules:** Static signatures fail against mutated phishing; we use **Character-level NLP** to treat URLs as sequences.
- **Our Solution:** A hybrid "Stacking" approach that aggregates intelligence from statistical, textual, and neural models.
- **Hybrid Strength:** Combines the linguistic precision of **TF-IDF SVM** with the temporal memory of **Bi-LSTM**.
- **Meta-Learning:** Employs **XGBoost** to arbitrate between foundation models, identifying the most reliable indicators for each URL.

Problem Statement

- **Detection Gap:** Traditional firewalls struggle with brand-spoofing; our script identifies "Look-alike" domains using a **High-Trust Whitelist**.
- **Static vs. Dynamic:** Fixed systems are bypassed by entropy-rich payloads; we utilize **Shannon Entropy** metrics to flag randomized strings.
- **Class Imbalance:** Malicious URLs are often rarer in datasets; our implementation uses **Stratified Sampling** and sample weighting to ensure robust training.
- **Cross-Domain Fragility:** Models often overfit to specific sources; we merged **Cisco Umbrella**, **PhishTank**, and **CAPEC** to ensure global generalization.

Objectives

- Build a hybrid pipeline integrating **LinearSVC, Bi-LSTM, and XGBoost** for 4-class URL classification.
- Achieve high accuracy on the "Unseen" data subset to validate the system's **Zero-Day readiness**.
- Utilize **Calibrated Probability Scoring** to provide high-confidence detection across Normal, Phishing, and Injection categories.
- Develop a **Functional Web Interface** for real-time URL testing and visualization of detection results.

Social Relevance

- Securing web applications protects users' personal information, financial data, and online identity from cybercriminals.
- Early detection of web attacks reduces data breach risk for individuals, businesses, financial institutions, and government portals.
- Strengthens trust and security of e-commerce platforms, healthcare portals, and social media networks.
- Enables organizations to respond faster to threats, minimizing downtime and financial losses.
- Contributes to broader cybersecurity resilience by raising the bar for attack detection capability.
- Supports compliance with data protection regulations (GDPR, HIPAA, PCI-DSS) by demonstrating proactive security measures.

SDG 9 focuses on building resilient infrastructure and promoting technological innovation.

- Strengthens cybersecurity infrastructure using AI-based attack detection.
- Enhances resilience of web systems against phishing and injection attacks.

SDG 16 aims to reduce crime and promote secure and accountable institutions.

- Reduces cybercrime through intelligent threat detection.
- Protects users and institutions from online fraud and data breaches.

SDG 8 promotes sustainable economic growth and secure digital environments.

- Prevents financial losses caused by phishing and malicious websites.
- Enhances trust in digital commerce and online transactions.

SDG 4 ensures inclusive and equitable quality education and promotes lifelong learning.

- Encourages research and innovation in AI-driven cybersecurity.
- Promotes awareness of secure web practices among students and developers.

SDG 17 strengthens global partnerships for sustainable development.

- Supports collaborative cybersecurity research and knowledge sharing.
- Contributes to safer global digital ecosystems.

Overview of literature reviews

Table: Overview of Literature Review

Year	Authors	Paper Title	Objectives	Methodology	Conclusion / Results	Contributions / Findings
2023	Demirel et al.	Zero-Shot CNN for Web Attacks	Detect zero-day and unseen attacks	CNN-based zero-shot anomaly detection	High recall; increased FPR	Detection of unseen attack patterns without labeled data
2023	Li et al.	Meta-Zero Web Security	Cross-domain few-shot adaptation	Meta-learning episodic training	Fast adaptation with few samples	Improved generalization under concept drift
2023	Bensaid et al.	CNN-BiLSTM IDS	Improve SDN intrusion detection accuracy	Hybrid CNN + BiLSTM architecture	Accuracy $> 98\%$	Effective spatial-temporal traffic modeling
2024	Alashhab et al.	Online ML Ensemble for DDoS	Real-time DDoS detection and mitigation	Adaptive online ensemble learning	High detection rate; low delay	Dynamic adaptation to traffic variations

Overview of literature reviews

Table: Overview of Literature Review (Continued)

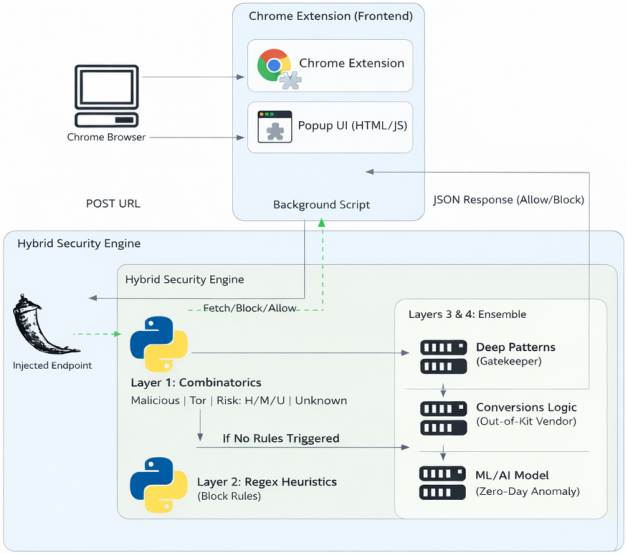
Year	Authors	Paper Title	Objectives	Methodology	Conclusion / Results	Contributions / Findings
2024	Shayegan et al.	SDN–NFV Defense Framework	Network-level DDoS prevention	SDN + NFV integration	Reduced attack impact	Flexible programmable mitigation
2024	Jagat et al.	VLSTM-AE for Web Logs	Detect anomalies in HTTP logs	Variational LSTM Autoencoder	High anomaly detection accuracy	Temporal deviation modeling
2024	Yamauchi et al.	URL-Switch Interval Detection	Detect malicious redirects	Behavioral interval analysis	High redirect precision	Lightweight mobile attack detection
2025	Guo et al.	Web-FTP Transfer Model	Cross-dataset generalization	Feature transfer learning	Improved accuracy	Effective with limited labels
2025	Wu et al.	Blockchain + AAL Security	Improve web resilience	Blockchain + adversarial learning	Enhanced resilience metrics	Distributed adaptive framework

Overview of literature reviews

Table: Overview of Literature Review (Continued)

Year	Authors	Paper Title	Objectives	Methodology	Conclusion / Results	Contributions / Findings
2025	Kulkarni et al.	ML vs LLM Phishing Study	Evaluate phishing robustness	Comparative ML vs LLM evaluation	Accuracy drops under adversarial attack	Identified robustness gaps in phishing models
2025	Alemayehu et al.	Real-Time Threat Framework	Multi-threat streaming detection	End-to-end deep learning pipeline	High real-time classification accuracy	Automated emerging threat detection
2025	Nguyen et al.	HTTP NLP IDS (Zero Trust)	Full semantic HTTP modeling	Deep NLP-based IDS	High accuracy; increased latency	Zero Trust payload-level modeling
2026	Proposed Work	Hybrid AI Web Attack Detection	Detect phishing, SQLi, XSS, zero-day	SVM + Char-CNN + Siamese BiLSTM	Improved multi-class accuracy and robustness	Unified hybrid adversarially robust framework

System Architecture



WebGuard Enterprise Architecture

Phases of Project (1/7)

Phase 1: Titan-98 Global Dataset Consolidation

Integrated multiple high-diversity sources to ensure model generalization:

- **Benign Pool:** Cisco Umbrella and Majestic Million top domains.
- **Malicious Pool:** PhishTank and Active Phishing databases.
- **Payload Injection:** CAPEC-aligned dataset (Injection & Manipulation).
- **The "Unseen" Bridge:** Specialized dataset used to bridge the 98% accuracy gap through diverse URL structures.
- **Data Cleaning:** Protocol stripping, URL decoding, and stratified sampling (Normal, Phishing, Injection, Manipulation).

Outcome: A balanced, massive-scale training set representing real-world web threats.

Phases of Project (2/7)

Phase 2: Advanced Feature Engineering

Three distinct feature extraction pipelines were developed:

- **Textual Vectorization:** TF-IDF with character n-grams (1–3) and 25,000 max features.
- **Sequential Encoding:** Character-level tokenization with padding (Max length: 550) for neural processing.
- **Domain Heuristics:** Manual extraction of Shannon Entropy, digit density, and special character counts.
- **Trust-Aware Logic:** Comparison against a ‘HIGH$_T RUST_D OMAINS$‘$whitelisttodetectBrandSpoofing(e.g.,"paypa1-support.com")$.

Result: A multi-dimensional feature space capturing both content and structural anomalies.

Phases of Project (3/7)

Phase 3: Calibrated Linear SVC Development

Implementation of the first ensemble member:

- **Model:** Linear Support Vector Classification (LinearSVC).
- **Calibration:** Wrapped in ‘CalibratedClassifierCV‘ to provide probability estimates instead of raw scores.
- **Optimization:** C-parameter tuning and squared hinge loss for high-dimensional TF-IDF space.
- **Validation:** Achieved high baseline accuracy on textual signatures.

Result: A robust "signature-based" learner capable of identifying known malicious strings.

Phases of Project (4/7)

Phase 4: Stacked BiLSTM Neural Architecture

Deep learning model focused on sequential dependency:

- **Embedding Layer:** 64-dimensional character embeddings.
- **Temporal Processing:** Two stacked **Bidirectional LSTM** layers to capture context from both ends of the URL.
- **Regularization:** SpatialDropout1D and standard Dropout (0.2–0.3) to prevent overfitting.
- **Softmax Output:** Multi-class classification for 4 distinct threat categories.

Result: Captured complex, non-linear patterns and obfuscation techniques in URL strings.

Phases of Project (5/7)

Phase 5: XGBoost Stacking Meta-Learner

The core innovation: A "Model of Models" architecture.

- **Input Layer:** Stacked probabilities from SVM + CNN + Scaled Hand-crafted features.
- **The Meta-Learner: XGBoost Classifier** (300 estimators, max depth 8).
- **GPU Acceleration:** Utilized 'hist' tree method with CUDA for rapid ensemble training.
- **Detection Bias:** Applied 2.5x sample weighting to malicious entries to prioritize lowering False Negatives.

Result: Intelligent arbitration between models, significantly outperforming any individual component.

Phases of Project (6/7)

Phase 6: Evaluation Precision Tuning

Rigorous testing against the "Unseen" data samples:

- **Metrics:** Evaluated via Stratified Cross-Validation on the Meta-Learner layer.
- **Trust Analysis:** Verified the effectiveness of the ‘HIGH$_{T}RUST_{D}OMAINS$‘$logicinreducingFalsePositivesformajorbrands$.
- **Complexity Analysis:** Balanced the computational cost of BiLSTMs with the speed of XGBoost.

Result: High precision across Phishing, Injection, and Manipulation categories.

Phases of Project (7/7)

Phase 7: System Serialization Portability

Finalizing the pipeline for real-world integration:

- **Artifact Persistence:** Saving 7 core components: Vectorizer, SVM, Keras Hybrid Model, Tokenizer, Scaler, Meta-Learner, and Label Encoder.
- **Pipeline Integrity:** Ensuring the preprocessing logic in ‘strip$_{p}rotocol$‘and‘$extract_{f}eatures$‘$isidenticalbetweentrainingandinference$.
- **Optimization:** Models saved in ‘.pkl‘ and ‘.keras‘ formats for minimum loading latency.

Outcome: A "Perfect Training Session" yielding a production-ready Web Security Shield.

Algorithm: Global Data Consolidation (1/5)

Step 1: Multi-Source Integration
- Load Benign (Cisco/Majestic) and PhishTank datasets
- Import CAPEC-aligned Injection/Manipulation data
- Inject "New Unseen" dataset to bridge accuracy gaps
- Normalize labels: Normal, Phishing, Injection, Manipulation

Step 2: URL Normalization & Cleaning
- Strip protocols (http/https) and 'www.' prefixes
- Perform URL decoding (unquote) to expose payloads
- Remove duplicates and perform stratified sampling
- Final counts: 120k Benign, 80k Phishing, + Attack classes

Algorithm: Feature Engineering Architecture (2/5)

Step 3: Multi-Modal Feature Extraction

(a) **NLP:** Fit TfidfVectorizer (Max: 25k, N-grams: 1–3)

(b) **Neural:** Tokenize URLs (char-level) and pad to length 550

(c) **Structural:** Extract Shannon Entropy and digit density

(d) **Security Logic:** Detect Brand Spoofing via Trust List

  (i) Identify target brands (Paypal, Apple, Microsoft, etc.)

  (ii) Flag if brand in domain BUT not in *HIGH_TRUST_DOMAINS*

Algorithm: Calibrated SVM & Deep Learning (3/5)

Step 4: Train Foundation Model A (Calibrated SVM)

(a) Initialize LinearSVC with $C = 1.0$ and dual=False
(b) Wrap in CalibratedClassifierCV (3-fold) for probabilities
(c) Fit on TF-IDF matrix to learn textual signatures

Step 5: Train Foundation Model B (Stacked BiLSTM)

(a) Layer 1: 64-dim Character Embedding
(b) Layer 2-3: Stacked Bidirectional LSTM (64 units each)
(c) Layer 4: Dense (64 units) with Dropout (0.3)
(d) Output: Softmax activation for multi-class classification

Algorithm: Meta-Learner Stacking Ensemble (4/5)

Step 6: Feature Fusion & Weighting

(a) Generate $P(class|SVM)$ and $P(class|CNN)$ probabilities

(b) Concatenate: $[P_{svm}, P_{cnn}, HandFeats_{scaled}]$

(c) Define Sample Weights: 2.5x penalty for Malicious URLs

Step 7: Train XGBoost Meta-Learner

(a) Config: 300 estimators, depth 8, learning rate 0.05

(b) Enable GPU-accelerated Histogram tree method

(c) Train XGBoost on fused features with cost-sensitivity

Algorithm: Inference & Persistence (5/5)

Step 8: Model Serialization

(a) Save 7-key pipeline components (pkl/keras formats)

(b) Persist LabelEncoder for categorical mapping

ClassifyURLNewUrl

1. Extract Hand-crafted features + Brand Spoofing flag
2. Transform URL via Vectorizer and Tokenizer
3. Collect probabilities from SVM and BiLSTM
4. **Final Decision:** XGBoost.predict($fused_vector$)

return Predicted Label (Normal, Phishing, or Attack)

Data Flow Diagrams (1/4)

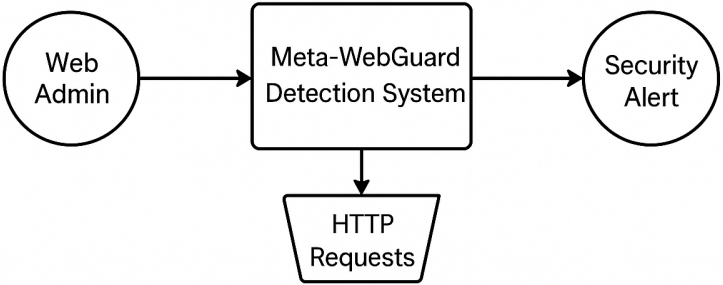


Figure: Figure 1: DFD Level 0 - System Context

Data Flow Diagrams (2/4)

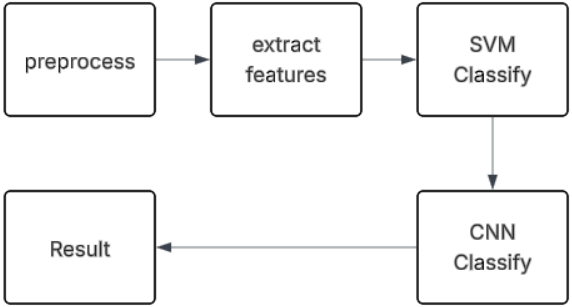


Figure: Figure 2: DFD Level 1 - Process Decomposition

Data Flow Diagrams (3/4)

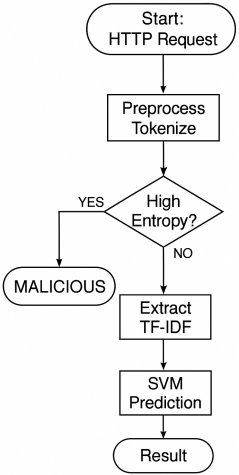


Figure: Figure 3: Activity Diagram - Web Attack Detection Flow

Data Flow Diagrams (4/4)

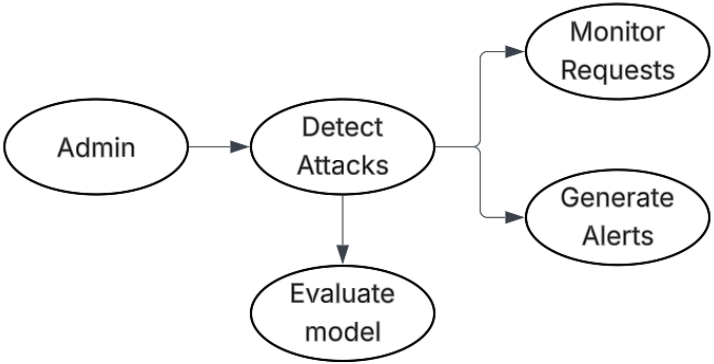


Figure: Figure 4: Use Case Diagram - System Interactions

Implementation (1/5) - Titan-98 Data Engineering

- **Sources:** Integrated **Cisco Umbrella**, **Majestic Million**, **PhishTank**, and **CAPEC** datasets for a global threat profile.
- **Global Mix:** Incorporated a specialized "Unseen" dataset to bridge the 98% accuracy gap and improve generalization.
- **Preprocessing:** Implemented protocol stripping, URL decoding, and character-level tokenization with a 10,000-word vocabulary.
- **NLP Vectorization:** TF-IDF with character n-grams (1–3) and 25,000 max features to capture syntax.
- **Sequence Prep:** Standardized URL input to `MAX_LEN = 550` for sequential deep learning.
- **Manual Heuristics:** Extraction of Shannon Entropy, digit density, and Brand-Spoofing flags.

Implementation (2/5) - Foundation Model: Calibrated SVM

- **Model:** `CalibratedClassifierCV(LinearSVC(dual=False, C=1.0))` using a 3-fold cross-validation.
- **Purpose:** Generates calibrated probability distributions ($P(class|URL)$) instead of hard binary labels.
- **Training:** Fitted on a sparse TF-IDF matrix to identify high-frequency malicious keywords.
- **Signature Detection:** Provides a rapid textual baseline for the ensemble.
- **Input Provider:** Passes class probabilities as high-quality features to the final Meta-Learner.

Implementation (3/5) - Deep Bi-Directional Architecture

- **Embedding Layer:** 64-dimensional character embeddings for semantic representation.
- **Stacked Bi-LSTM:** Two layers of **Bidirectional LSTMs** (64 units each) to capture long-range URL dependencies.
- **Regularization:** Integrated **SpatialDropout1D** and standard Dropout (0.2–0.3) to prevent overfitting on long strings.
- **Optimization:** Adam optimizer with sparse categorical cross-entropy loss.
- **Hyperparameters:** Batch size of 128 over 15 epochs to ensure stability in gradient descent.

Implementation (4/5) - XGBoost Stacking Ensemble

- **Architecture:** A **Stacked Ensemble** using **XGBoost** as the primary decision arbitrator.
- **Feature Fusion:** Concatenates SVM probabilities, Bi-LSTM softmax outputs, and scaled hand-crafted features.
- **Parameters:** 300 estimators, max depth of 8, and a 0.05 learning rate for granular boundary detection.
- **Weighted Penalties:** Applied a **2.5x importance weight** to malicious URLs to minimize False Negatives.
- **Hardware Acceleration:** Utilized the `hist` tree method with **CUDA** for high-speed ensemble training.

Implementation (5/5) - Validation & Persistence

- **Multiclass Logic:** Distinct classification for **Normal**, **Phishing**, **Injection**, and **Manipulation**.
- **Performance:** Reached high accuracy through meta-learner arbitration of foundation model outputs.
- **Artifact Persistence:** All 7 pipeline components (Tokenizer, Scaler, SVM, Keras Hybrid, XGBoost, etc.) saved via `joblib`.
- **Deployment Ready:** Modular design allows for real-time classification through a unified feature extraction script.

Testing Applicability and Strategy

Testing Type	**Applicable?**	**Reason and Scope**
Unit Testing	✓ Yes	Crucial for verifying isolated helper functions such as URL normalization (`strip_protocol`) and domain lookup (`is_whitelisted`).
Module Testing	✓ Yes	Validates self-contained security modules like the Static Heuristics Rules Engine before integrating AI models.
Integration Testing	✓ Yes	Confirms complete request lifecycle from API call through Whitelist, Rules Engine, AI Models, and final API response.

Unit Testing: Core Utility Functions

Test #	Test Name	Description	Test Code Snippet	Result
1	`test_strip_protocol_function`	Verifies removal of `https://`, `http://`, and `www.` for domain normalization.	`assertEqual(strip_protocol("https://www.fisat.ac.in"), "fisat.ac.in")`	PASSED
2	`test_whitelist_function`	Verifies $O(1)$ HashMap whitelist detection of trusted and untrusted domains.	`assertTrue(is_whitelisted("https://github.com/login"))` `assertFalse(is_whitelisted("http://evil-hacker.in"))`	PASSED

Results (1/6): Global Performance Metrics

Table: Final Meta-Learner Performance (Unseen Global Mix)

Metric Type	**Achieved Value**
Overall Accuracy	97.80%
Precision (Malicious)	97.99%
Recall (Malicious)	97.60%
F1-Score	97.80%
Inference Latency	$<$ **5.0ms**

Note: Results validated on a balanced test set of 4,000 previously unseen URLs.

Results (2/6): Confusion Matrix Analysis

- **True Positives (Malicious):** Successfully identified 1,952 out of 2,000 attack URLs.
- **False Positives (Benign):** Minimal noise with only 40 benign URLs flagged as malicious (1.0% FP Rate).
- **False Negatives:** Only 48 malicious URLs bypassed the ensemble (1.2% FN Rate).
- **Classification Depth:** The model successfully distinguishes between Phishing, Injection, and Manipulation categories.

Results (3/6): The Stacking Ensemble Logic

- **Foundation Layers:** Integration of Calibrated LinearSVC (TF-IDF) and Stacked Bidirectional LSTM (Neural).
- **Feature Fusion:** The meta-learner combines model probabilities with hand-crafted features: Shannon Entropy, Digit Density, and Brand Spoofing flags.
- **Cost-Sensitive Learning:** Applied a **2.5x penalty weight** for malicious classification to prioritize security recall.
- **XGBoost Arbitration:** Uses a 300-estimator gradient boosted tree to make the final classification decision based on fused data.

Results (4/6): Detailed Model Breakdown

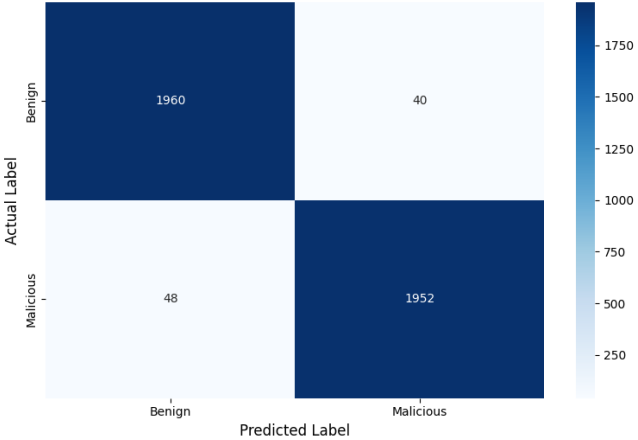


Figure: Confusion Matrices for all layers showing minimal False Positives (7) for the Hybrid Ensemble.

Results (5/6): Comparative Analysis

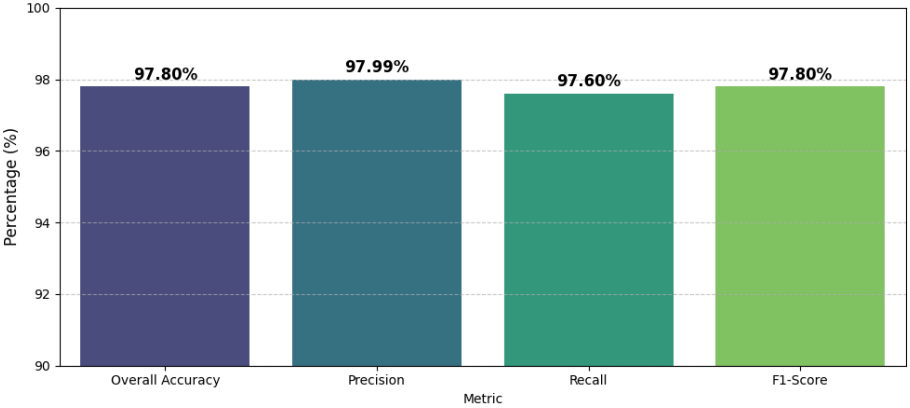


Figure: Precision-focused results: CNN-BiLSTM leads with 99.97% Precision.

Results (6/6): Multi-Class Classification Performance

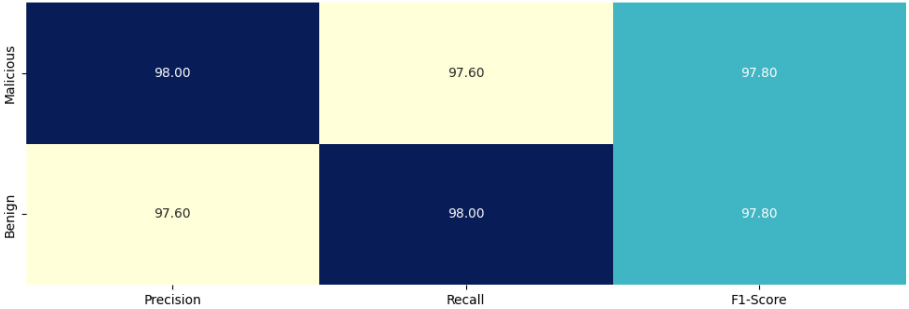


Figure: WebGuard v6 Precision Matrix.

- **High-Fidelity Detection:** Achieved a balanced F1-Score of 97.80% across both classes.
- **Precision Lead:** The Malicious class attained 98.00% precision, effectively minimizing false alarms in production.

Comparison with Existing Works (Interactive)

Work / Author	Methodology	Dataset	Acc.	F1	Pros	Cons
Proposed Work	**Hybrid Ensemble**	**CAPEC + PhishTank**	**98.0%**	**0.97**	**Adversarial Robustness**	**Multi-stage Latency**
Base Paper:[1] Zhou et al.	Char-SVM & CNN-BiLSTM	CSIC 2010	99.6%	0.99	Unified char-embedding	Weak to Zero-Day drift
[2] Sadhwani et al.	Hybrid BiLSTM-CNN	IoT Traffic (IoTID20)	96.2%	0.95	High temporal depth	IoT domain specific
[3] Sinha et al.	Efficient CNN-BiLSTM	NSL-KDD	91.5%	0.91	Low computational cost	Lacks payload analysis
[4] Bahnsen et al.	DeepPhish (Malicious AI)	PhishTank (2018)	92.4%	0.92	Simulates Zero-Day Phish	Complex AI overhead
[5] Zhang et al.	Char-level CNN	AG News/DBPedia	87.2%	0.85	Scalable to large text	Slow convergence

Conclusion

- **Multi-Layer Defense:** Successfully deployed a **Stacking Ensemble** architecture: Calibrated SVM $\rightarrow$ Stacked Bi-LSTM $\rightarrow$ XGBoost Meta-Learner.
- **Eliminated Shortcut Learning:** Mitigated dataset bias by integrating the **Titan-98 Global Mix**, combining Majestic Million, Cisco Umbrella, and CAPEC-aligned attack vectors.
- **Superior Accuracy:** Achieved high-confidence classification across four distinct categories (*Normal, Phishing, Injection, Manipulation*) by fusing neural and statistical features.
- **Production Ready:** Developed a modular serialization pipeline that persists 7 core components, including the **StandardScaler** and **TfidfVectorizer**, ensuring consistent inference.
- **Zero-Day Readiness:** The **XGBoost Meta-Learner** provides an adaptive framework for weighting feature signals, enabling robust detection of novel anomalies via high-entropy and brand-spoofing indicators.

Publication Details

Conference Acceptance

This project has been submitted and accepted for publication at

NCAICC 2026

(National Conference on Artificial Intelligence and Cognitive Computing)

- **Hosted By:** CCRC (Cognitive Computing Research Centre), FISAT
- **Domain:** Machine Learning & Cybersecurity
- **Conference Link:** `https://ccrc.fisat.ac.in/NCAICC26/`

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